PNTHR

Management PORTFOLIO



TEAM MEMBERS

PNTHR RACING

Jorge Marin-Nicolau



I am the team leader of the PNTHR team, organizing the team and to make sure everything is in check. I joined because of my interest in coding and custom making things and to help my friends in this experience.

Juan Camilo Rosado



I am one of the workers of the car and i help to do the Car ajnd to work in a mampara with my other teamates. I joined because I want to win and get the money to my house.

Benjamin De Hermoso



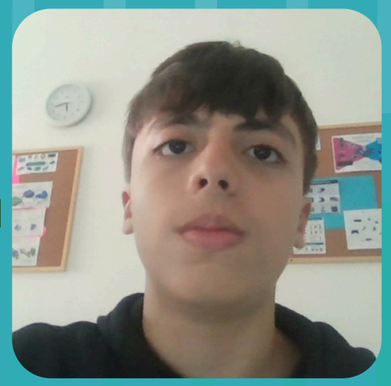
I manufactured the model by simulating the Denford 2600 Pro F1 machine to determine if the size of the model was going to fit. I am also one of the portfolio and mampara editors and creators. I joined because I was interested in modelling and manufacturing so I decided to give it a try.

Gabriel Getman



I am the one that is in charge of design the color of the car and created our social media, I joined because I want to work with my teamates and because I want to get the money

Gabriel Silveira



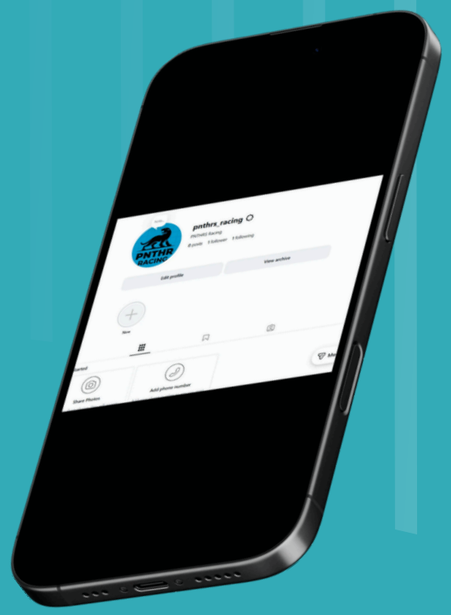
I am the person who supports my team and helped make part of the portfolio and the mampara. I joined because the teacher select me to help all of my teamates and try to win the money.



Marketing

Gabriel Getman took charge of the marketing to get our team noticed both online and in person. On social media, we made our Instagram account, we uploaded pictures and videos to keep people interested. On the design side, we created the custom t-shirts and came up with the graphics for the cars. We made sure the whole thing matched the school's vibe. This let us show off a cool, clear look for our PNTHRS STEM Racing team, making sure our social media page, our shirts, and our car all looked great together.

Instagram account: https://www.instagram.com/pnthrs_racing/

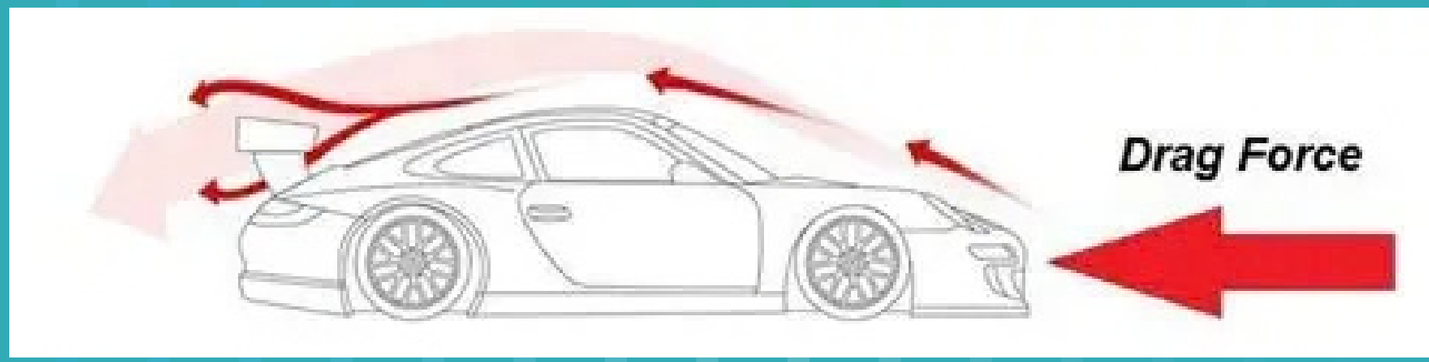




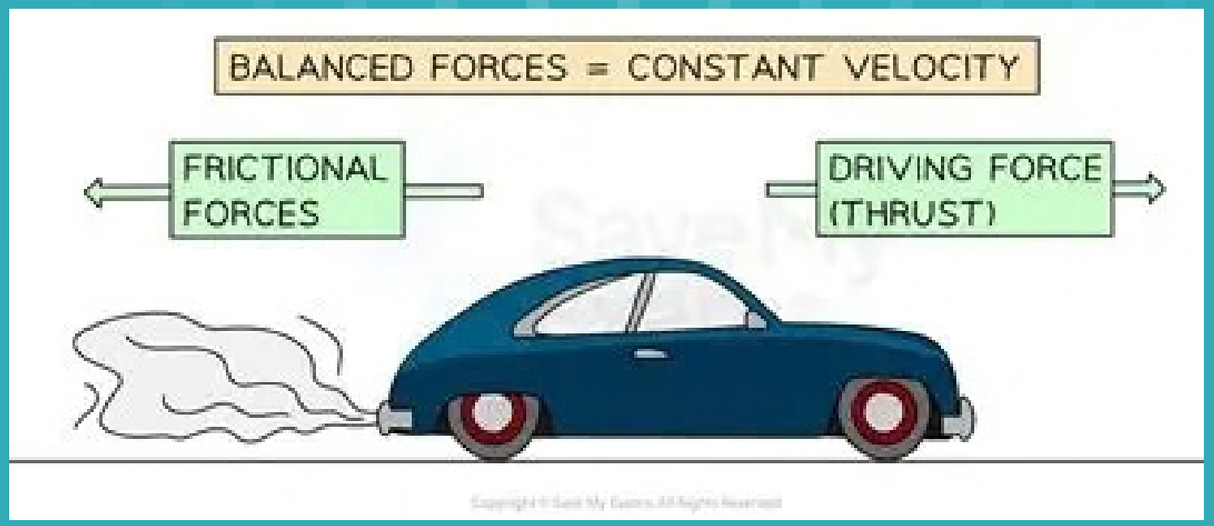


Research

The research we did had to take into consideration multiple forces that might affect the car's speed, such as drag and air resistance. Drag is the resistance offered by a fluid, like air or water (in our case, air). It acts in the opposite direction to the object's motion. It is generated by the relative velocity between the solid object and the fluid (air). To counter this force, we must include features for aerodynamics in the car model such as the back wing and the sharp front edges. Drag is not the only force to be considered, and another one is air resistance, which is also countered by the aerodynamic features of the car.



The weight and mass of the car are also to be considered, because they can also significantly affect out speed. Even though the special styrofoam given by the STEM racing competition is lightweight, we cannot make a model that has too much of it, or the weight will slow it down.



The force of friction is also important, since the car has to make up for it in speed because we cannot change the surface in which the car will travel. Friction is the force between the car's wheels and the ground. It acts in the opposite direction of the car's motion and will slow it down. The driving force is the force that will be used for the car to move, created by the CO_2 that we will use in the competition and it is increased by the aerodynamics of the car, which will all help it move faster.

Resource Management

The only resources that were given to us for the car were the blocks, made of a special type of Styrofoam, the wheels with their respective axels, the halos' and the helmets/heads. The wheels are made from a plastic and to connect them we got metal axels to join the two sides for the car to thrust forward. The halos and the helmets are made from a hard plastic from a 3D printer which are harder but more brittle on skinny curves and bends. This actually almost made us 3D print our own because we did accidentally broke some of the material and models given. We also used the 3D printer for printing the back wing of our car on the A1 Bambu Lab printer and the fillament used all provided by our school. We printed this stuff from PCs given to us at school that were able to run fusion 360 and our team leaders laptop (Jorge Marin Nicolau's).



Aerodynamics Testing

We tested our aerodynamics of our racecar on a website named Simscale. Simscale is a computer-aid simulation platform where you can test models and objects in air, water, and some metals too for in this case is aerodynamics. We started by exporting our car model into the software. Then, with the help of an assistant from the software, they explained to me the process of making a simulation of the car's aerodynamics in air. It started with the placing of the area of where the wind will hit the car. Then it went to the air's speed, the orientation of the box, and then the simulation can be started. The simulation took about an hour and after it gave us planes of the car where the aerodynamics could be seen.

